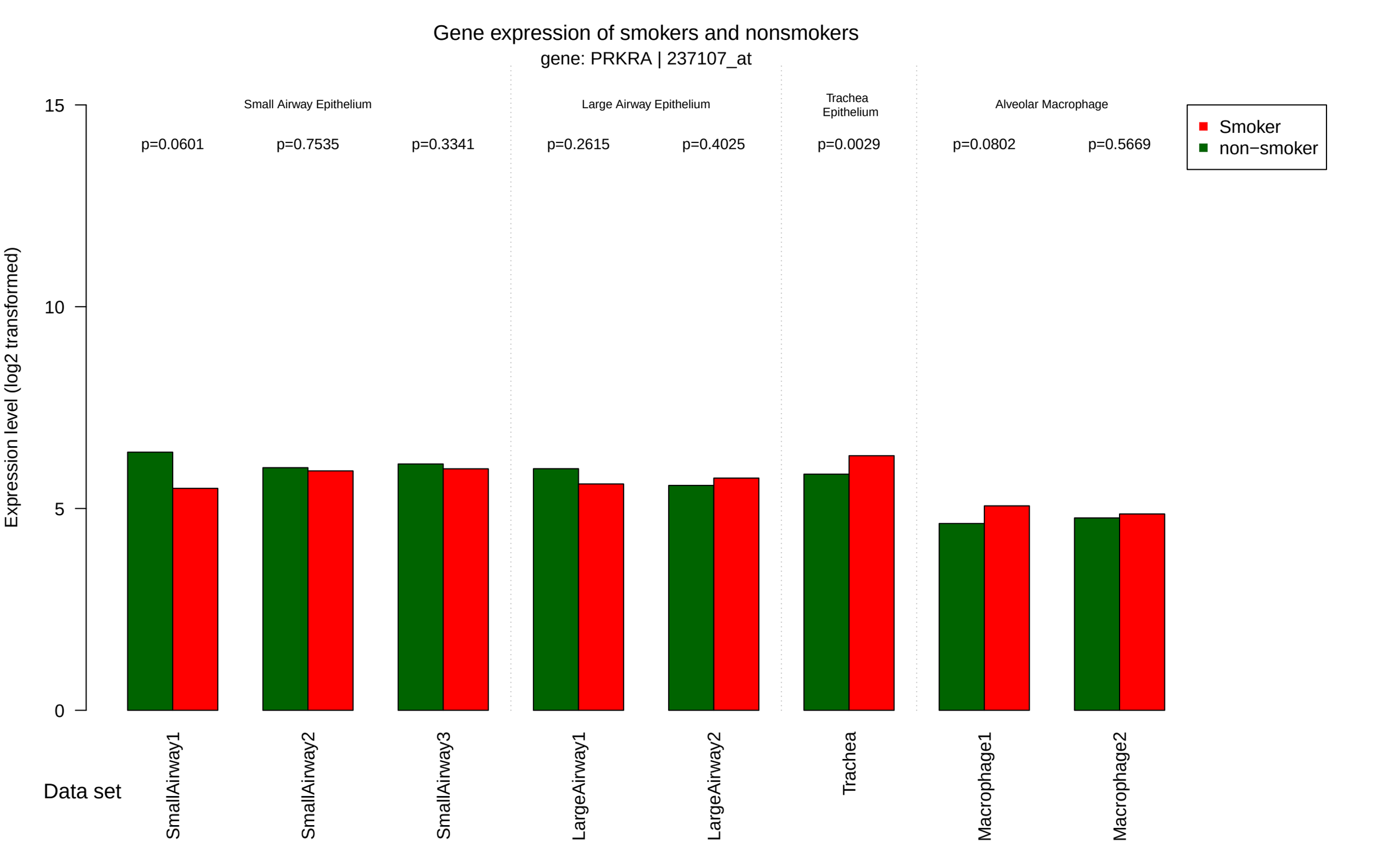
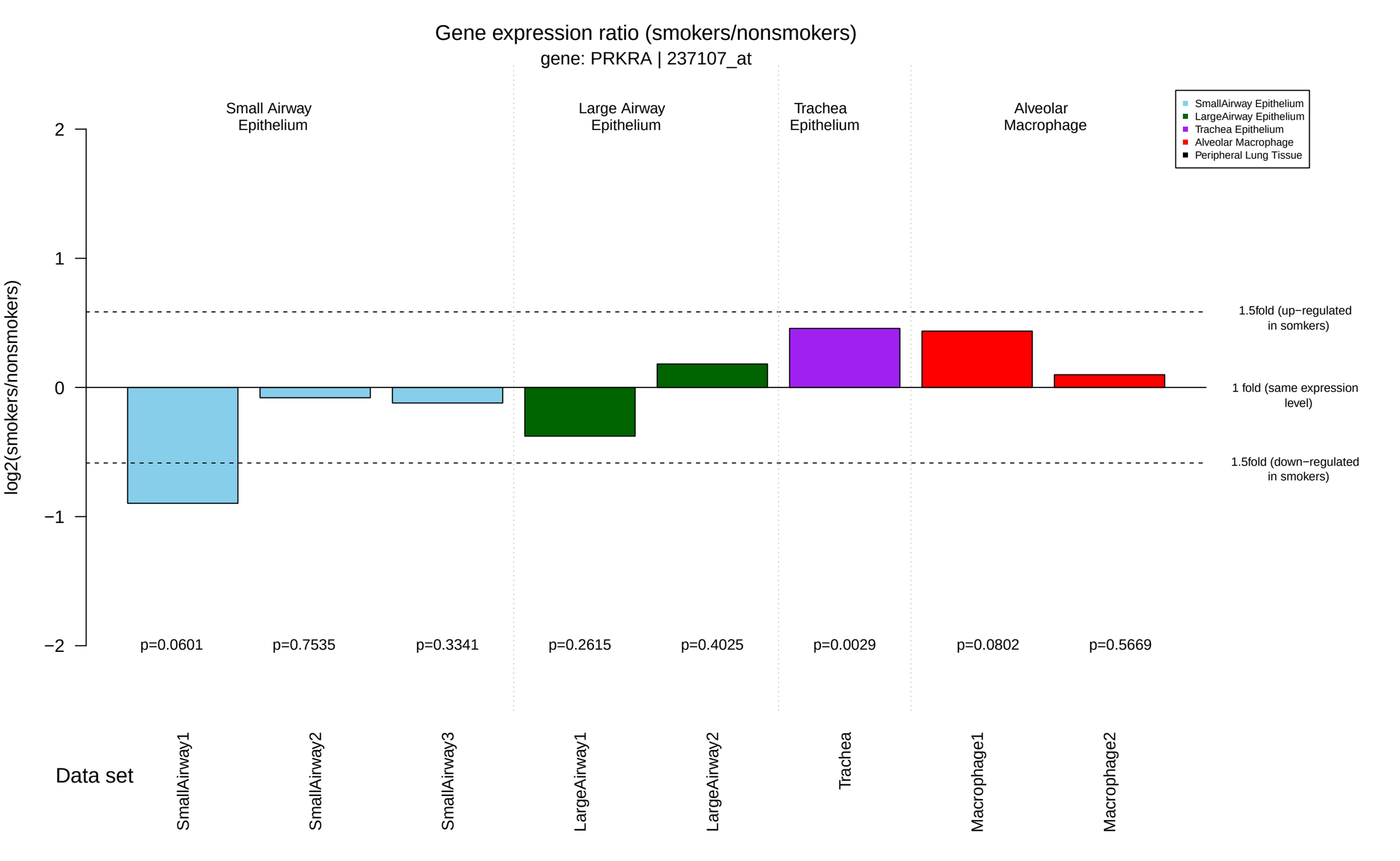
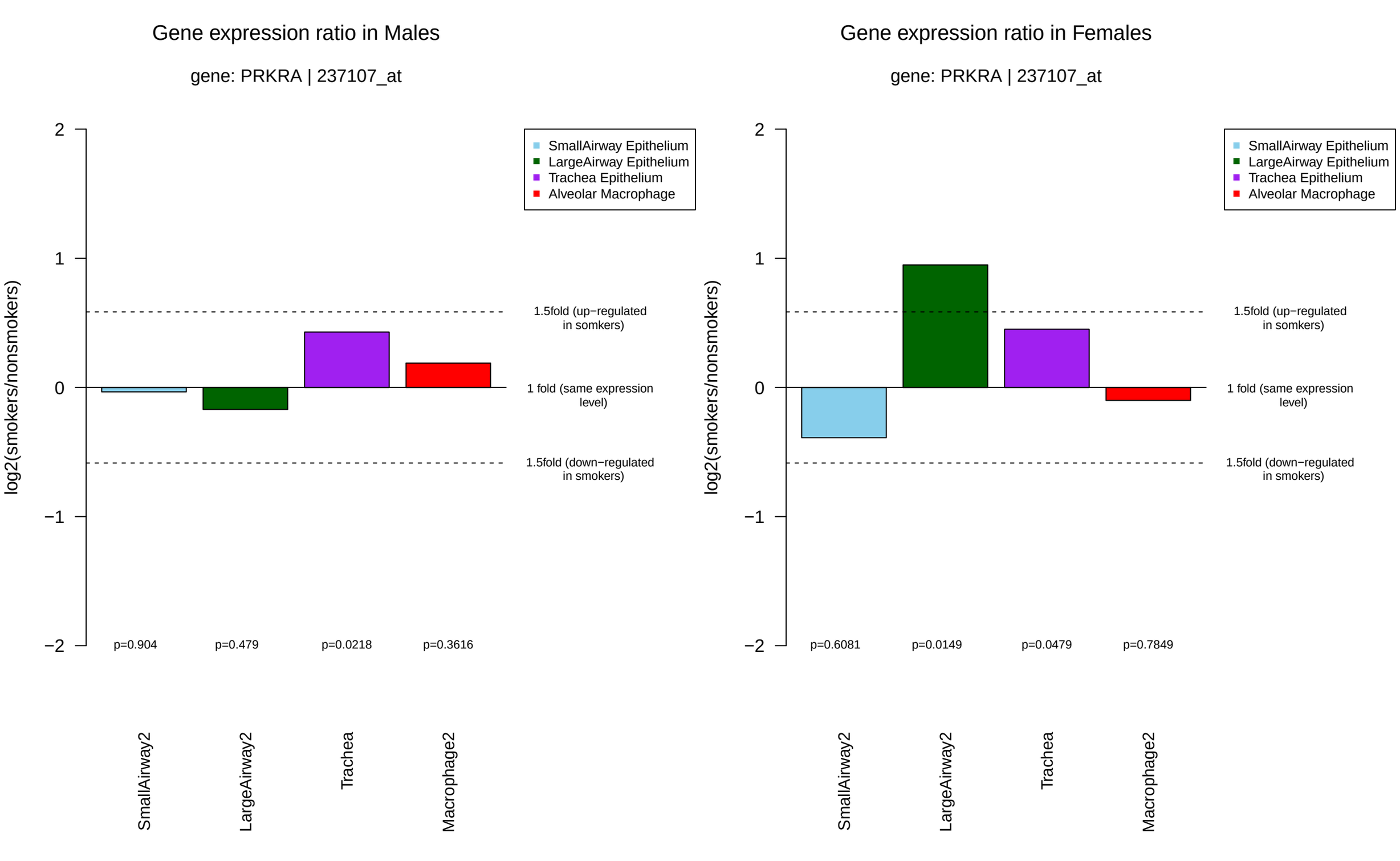
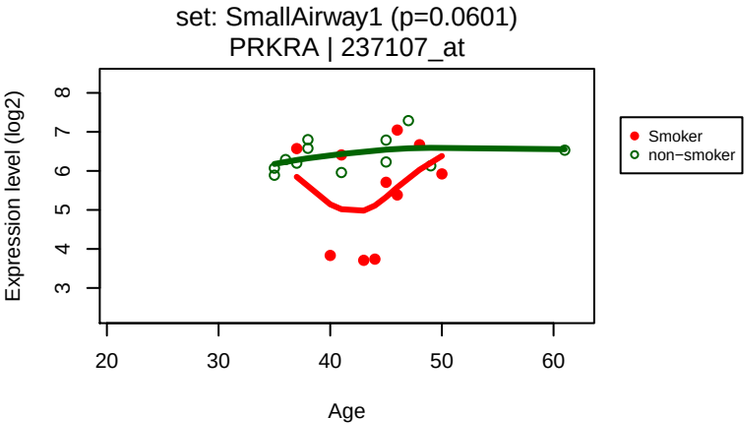
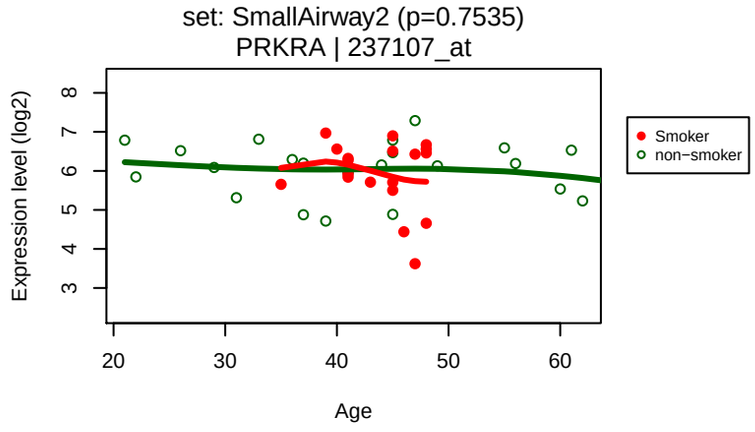
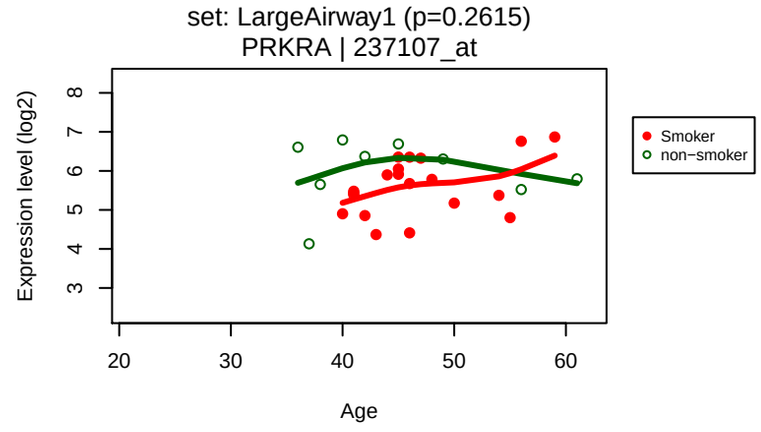

Gene expression of smokers and nonsmokers
gene: PRKRA | 237107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.7535
p=0.3341
p=0.2615
p=0.4025
p=0.0029
p=0.0802
p=0.5669
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

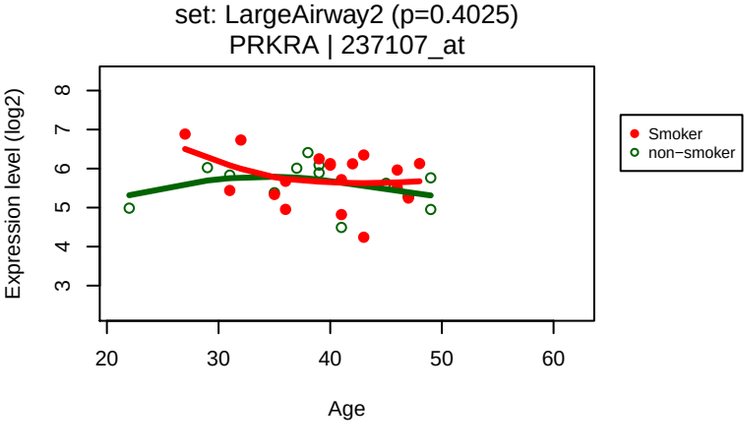

Gene expression ratio (smokers/nonsmokers)
gene: PRKRA | 237107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.7535
p=0.3341
p=0.2615
p=0.4025
p=0.0029
p=0.0802
p=0.5669
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

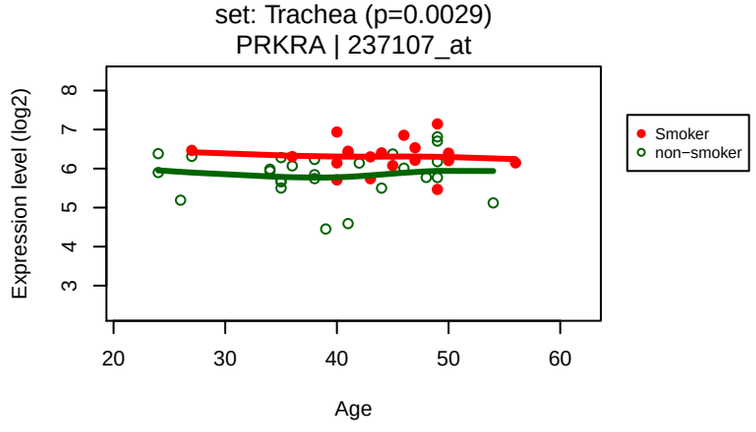

Gene expression ratio in Males
gene: PRKRA | 237107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.904
p=0.479
p=0.0218
p=0.3616
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKRA | 237107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6081
p=0.0149
p=0.0479
p=0.7849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

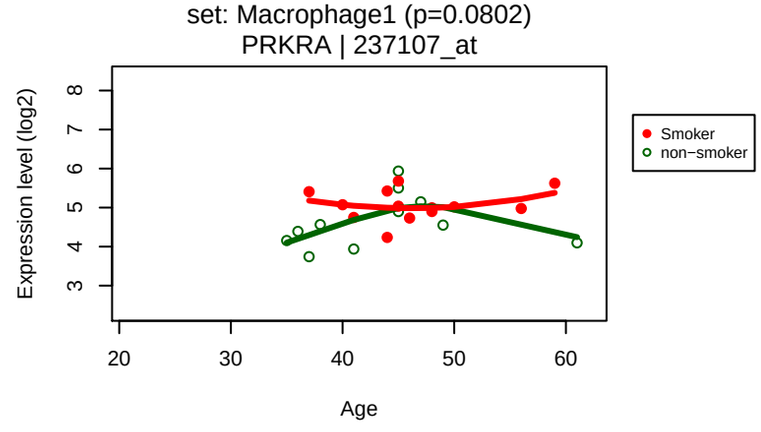

set: SmallAirway1 (p=0.0601)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker

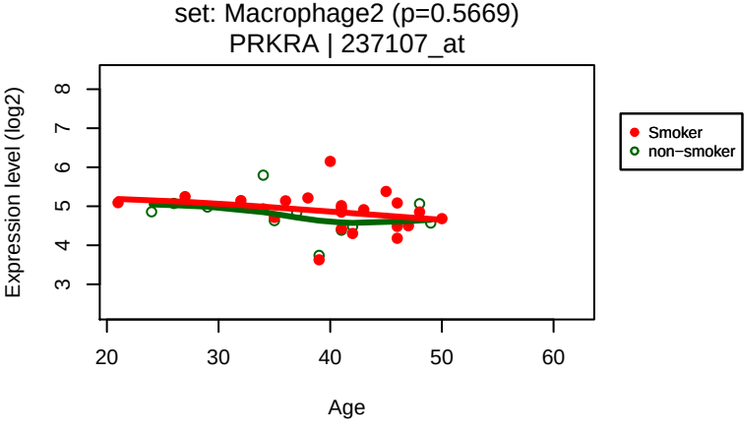

set: SmallAirway2 (p=0.7535)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker

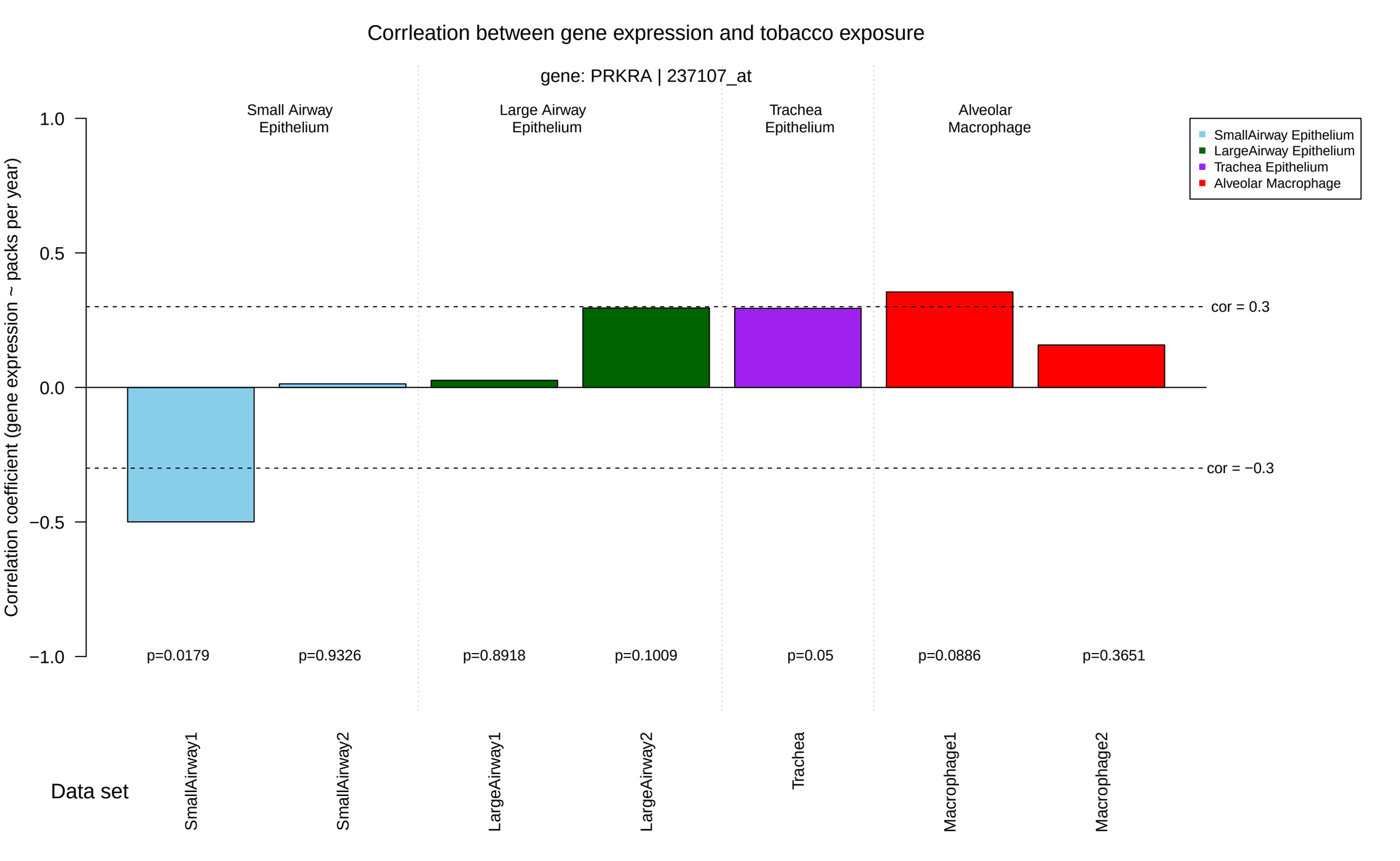

set: LargeAirway1 (p=0.2615)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4025)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0029)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0802)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5669)
PRKRA | 237107_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKRA | 237107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0179
p=0.9326
p=0.8918
p=0.1009
p=0.05
p=0.0886
p=0.3651
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2